

as alternative graders. The model includes predictors for the grader model, the investigated model, the organism type, the activation source (activation difference ($\bar{\delta}$), average base model activations ($\bar{\mathbf{h}}_{\text{base}}$), or average finetuned model activations ($\bar{\mathbf{h}}_{\text{ft}}$)), and the token projection method (Patchscope or Logit Lens). We exclude the position from which activations are read, as including it degraded model performance. The posterior feature effects for the token relevance grader are shown in Figure 9.

We observe clear differences between the graders: the posterior grader effects’ credible intervals don’t overlap with each other and zero. Notably, GPT 5 Mini results in more tokens being deemed relevant than the other graders, while Claude 4.5 Haiku results in the fewest tokens being deemed relevant. We evaluated Claude 4.5 Haiku with its default settings, which deactivates reasoning. This might explain the lower relevance scores. Additional effects we observe include that Patchscope results in more tokens being deemed relevant than Logit Lens. The source is clearly the strongest predictor of token relevance, with the activation difference being the strongest positive predictor—this aligns well with our findings. We observe that some model and organism-type effects are weak but credibly non-zero; for example, EM appears to result in slightly more tokens being deemed relevant than other organism types.

To assess inter-grader agreement, we compute Krippendorff’s α (Krippendorff, 2018) across the three grader models. For binary token relevance (nominal), we obtain $\alpha = 0.65$, indicating moderate but imperfect agreement between graders. This level of agreement is consistent with the systematic differences in grader behaviour revealed by the GLM analysis.

In Figure 11, we display the token relevance results with Patchscope for the different graders. We can see that for all graders, the observed result holds—the activation difference $\bar{\delta}$ results in many more tokens being deemed relevant than the other sources.

D.2 PATCHSCOPE SCALING FACTOR GRADER

We analyze how different grader models affect the patchscope scaling factor. As described in Appendix C.1, we ask the grader to select one of 30 scaling factors that produces the largest set of semantically coherent tokens. We rerun the patchscope scaling factor analysis for all organisms except the SDF organisms on the largest model Qwen3 32B, comparing Gemini 2.5 Flash (Comanici et al., 2025) and Claude 4.5 Haiku (Anthropic, 2025) as alternative graders. Similar to before, we fit a GLM using HiBayes (Luettgau et al., 2025; Dubois et al., 2025) with an ordered logistic likelihood. We run the study only on the activation differences $\bar{\delta}$. The posterior feature effects for the patchscope scaling factor grader are shown in Figure 10.

We do not find robust evidence for systematic differences between the grader models: the posterior grader effects’ credible intervals all include zero. The strongest predictor of the scaling factor is organism type, where Taboo results in a higher scaling factor than other organism types, while Subliminal results in a lower scaling factor. Since the grader model is not a significant predictor, we do not perform additional token relevance analysis with different patchscope scaling factor graders.

Figure 10 shows that we do not find robust evidence for systematic grader bias in the chosen patchscope scale: the posterior grader effects are close to zero and their 95% HDIs overlap the global mean. However, agreement on the exact scaling factor for individual items is only moderate. Pairwise Pearson correlations between the scalar scaling factors selected by the three graders range from $\rho = 0.46$ to $\rho = 0.65$ (all $p < 0.001$), and Krippendorff’s α (Krippendorff, 2018) with an ordinal distance metric on the index of the selected scale is $\alpha = 0.55$. This indicates that graders share substantial signal beyond chance but are not interchangeable: they broadly track each other, yet often disagree on the precise scaling factor. We hypothesise that this stems from the looseness of the instruction to “choose the factor with the most semantically coherent set of tokens,” which likely induces substantial noise. Given that our downstream agent nevertheless performs well and the token relevance metrics show a clear and robust signal, we consider this level of noise acceptable for the purposes of the present study. Developing a more precise and reliable scale-selection protocol is an interesting direction for future work, as well as evaluating whether the agent performs better with having access to multiple scales.

D.3 COHERENCE GRADER

We analyze how different grader models affect the maximal steering strength at which outputs remain semantically coherent. As described in Appendix C.3, for each organism we gradually increase the

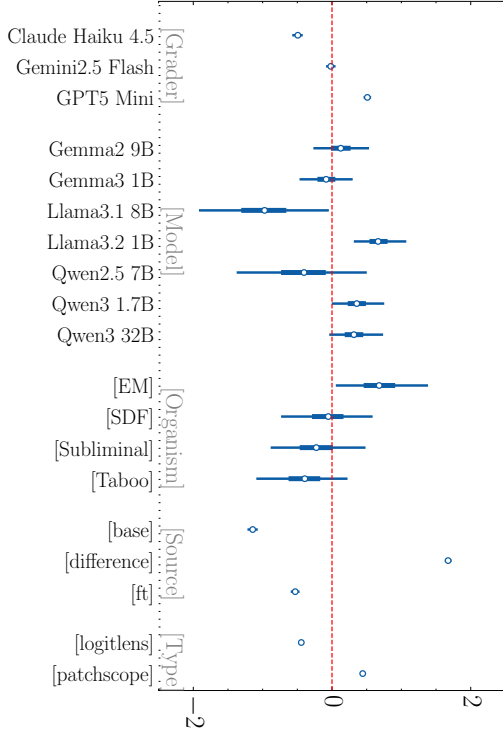


Figure 9: Posterior feature effects (x -axis) from a hierarchical GLM fitted using an ordered logistic likelihood for the **token relevance outcome**. Points show posterior point estimates of coefficients; vertical bars show 95% Highest Density Intervals (HDIs). Effects are parameterized for grader model, investigated model class, organism type, source of activations, and token projection method.

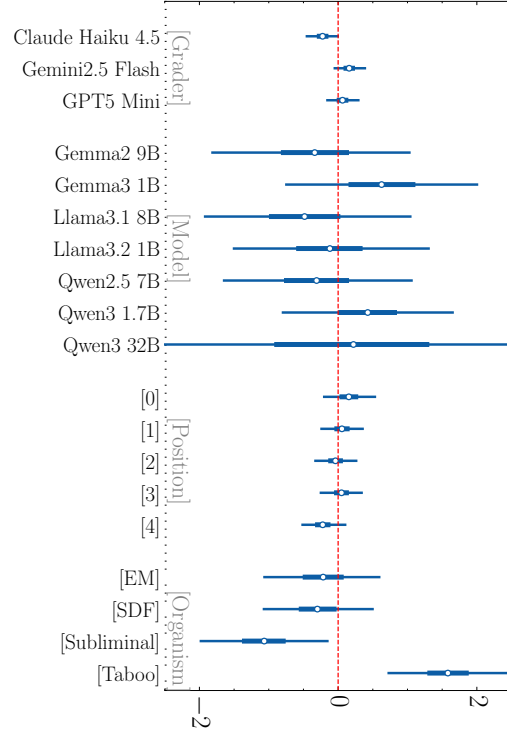
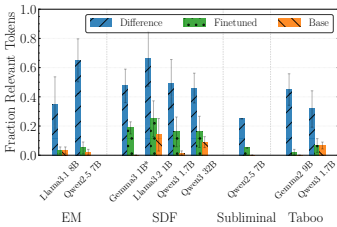
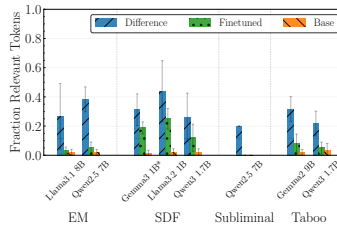


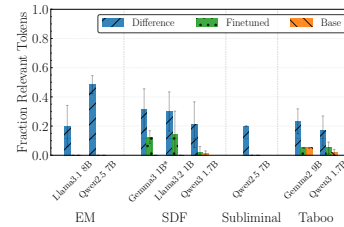
Figure 10: Posterior feature effects (x -axis) from a hierarchical GLM fitted using an ordered logistic likelihood for the **patchscope scaling factor outcome**. Points show posterior point estimates of coefficients; vertical bars show 95% Highest Density Intervals (HDIs). Effects are parameterized for grader model, investigated model class, extraction position and organism type.



(a) GPT 5 Mini



(b) Gemini 2.5 Flash



(c) Claude 4.5 Haiku

Figure 11: Token relevance grader ablation results comparing different language model graders (GPT 5 Mini, Gemini 2.5 Flash, and Claude 4.5 Haiku) on our bias detection pipeline. Each subplot shows the percentage of relevant tokens identified by each grader across different model organisms and configurations.

steering coefficient and repeatedly query a grader model, which returns a binary judgment of whether the resulting completion is still coherent. We then use a simple search procedure to identify the largest steering coefficient for which the grader still judges the output as coherent, and record this value as the *maximal safe steering strength* for that (organism, position, model, grader) configuration. In contrast to the discrete factors used elsewhere, this yields a continuous outcome defined on the underlying steering parameter. We repeat this process for a subset of the organisms with the grader model Gemini 2.5 Flash Lite (Comanici et al., 2025). The subset we use with the grader model Gemini 2.5 Flash Lite are the SDF organisms for the models Qwen3 1.7B and Llama3.2 1B Instruct as well as the EM organisms for the model Qwen3 32B.

To study how grader choice influences these maximal safe strengths while accounting for other factors, we fit a hierarchical GLM using HiBayes (Luettgau et al., 2025; Dubois et al., 2025). Let y_i denote the (log-transformed) maximal safe steering strength for configuration i . We initially attempted to model all factors (grader model, steered base model, intervention position, and organism type), but found that including position and organism type led to convergence issues. We therefore fit a reduced model with only grader and base model effects:

$$y_i \sim \mathcal{N}(\mu_i, \sigma_{\text{obs}}), \quad \mu_i = \alpha_0 + \alpha_{g(i)}^{(\text{grader})} + \alpha_{m(i)}^{(\text{model})}, \quad (3)$$

where $g(i), m(i)$ index the grader model and steered base model. Each family of effects has a hierarchical prior

$$\alpha_g^{(\text{grader})} \sim \mathcal{N}(0, \tau_{\text{grader}}), \quad \alpha_m^{(\text{model})} \sim \mathcal{N}(0, \tau_{\text{model}}), \quad (4)$$

with weakly informative half-normal priors on the scales τ . and on σ_{obs} . Categorical predictors are effect-coded, so all coefficients can be interpreted as deviations from the global mean α_0 . The posterior feature effects for the coherence grader are shown in Figure 12.

We observe meaningful differences between the graders: the posterior grader effects’ credible intervals do not overlap with each other or with zero. GPT 5 Nano yields slightly higher steering strengths than Gemini 2.5 Flash Lite. Beyond small systematic differences captured by the GLM, graders give very similar scores to individual items (pairwise Pearson correlations between graders is $\rho = 0.928, p < 0.001$). Further, the grader model effect is substantially weaker than the effect of the base model class being steered. Smaller models generally seem to result in lower steering strengths than larger models. Figure 14 displays the steering results for the different coherence graders. Across all graders, the key finding remains consistent: steering with the activation difference $\bar{\delta}$ produces much higher similarity to the finetuning dataset than other sources.

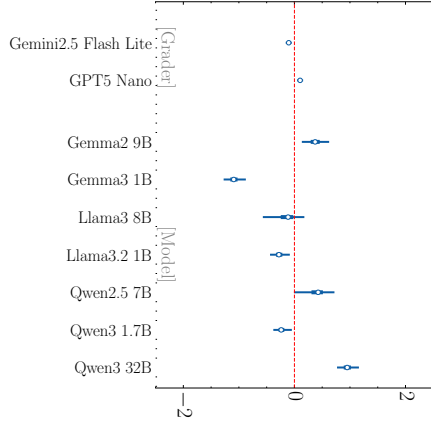


Figure 12: Posterior feature effects (x -axis) from a hierarchical GLM fitted using an ordered logistic likelihood for the **steering strength outcome**. Points show posterior means; vertical bars show 95% Highest Density Intervals (HDIs). Effects are parameterized for grader model and investigated model class.

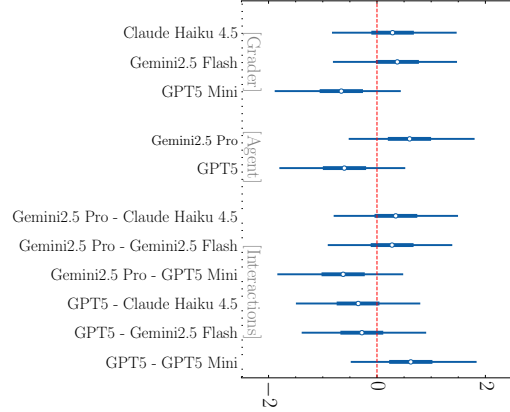
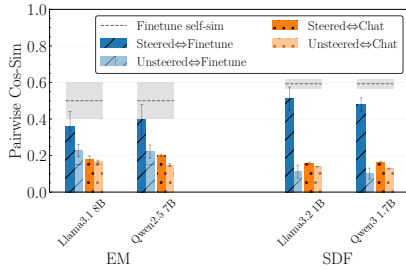
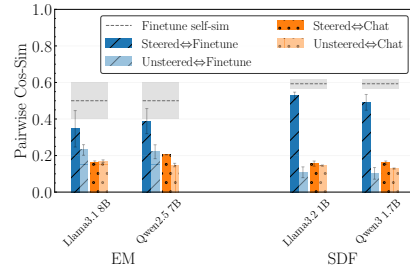


Figure 13: Posterior feature effects (x -axis) from a hierarchical GLM fitted using an ordered logistic likelihood for the **agent grades**. Points show posterior means; vertical bars show 95% Highest Density Intervals (HDIs). Effects are parameterized for grader model, agent model, their interactions, ADL access, investigated model class, and interaction budget i . Only the effects of the grader model, agent model, and their interactions are displayed.



(a) Gemini 2.5 Flash Lite



(b) GPT 5 Nano

Figure 14: Steering results for the different graders on the SDF organisms (only Qwen3 1.7B and Llama3.2 1B Instruct) and EM organisms. Average pairwise cosine similarity between text embeddings of steered outputs, unsteered outputs, the finetuning dataset, and normal chat data. The gray dotted line indicates within-finetuning-dataset cosine similarity, with the shaded area representing the standard deviation.